BRK - Autoloop

User manual and parameter reference · 1.2.1

A sample-folder rhythm instrument with deterministic random selection and per-step control lanes.

Quick start

1. Insert Autoloop on an instrument track, add a row and choose a folder containing your samples.
2. Enable steps in that row and start the host, or select its internal clock.
3. Add a lane, draw a small variation and try Evolve or a different seed. Save the host session to retain the pattern.

Editor window

1260 × 561 minimum; resizable, with scrolling for additional rows. Dimensions are logical editor pixels, before host window chrome. HELP adds 92 pixels of height (Gear adds a one-pixel divider too). Allow enough usable screen space at your display scale. Press HELP again to close the help area.

Operation

The host session recalls top-level parameters, rows, patterns, steps, lanes, locks, Evolve settings, seeds and sample references. Autoloop does not currently have a separate product preset manager.

HELP shows the explanation of the hovered or keyboard-focused control. The sequencer drawing and lanes still require a pointer. Folder/sample references are stored with relative paths and absolute fallbacks; copying the same sample folders to another user account helps projects travel.

How it works

- **Clock**: `Host` follows the DAW's song position (PPQ), so the pattern lands the same way on every playback and after every loop or jump; triggers fall on the exact sample of each step boundary. `Internal` runs its own clock from the BPM knob, so the plugin keeps playing with the transport stopped.
- **Rate**: the step length, from 1/4 to 1/32 with dotted and triplet values between (1/8T is three steps per beat, 1/16D is a step and a half), all still locked to the beat. A row has up to 32 steps; its `Steps` box in the Controls area (drag up or down, or double-click to type) sets how many it cycles through, 1 to 32 (16 by default), so a 3-step row against a 16-step row makes a polyrhythm. A row over 16 steps draws all its cells narrower. Cells past the length are drawn as empty outlines and are inert. Groove and Swing follow the 16-step grid of the song position; each row's Evolve cycle is its own length.
- **Groove and Swing**: `Groove` picks a timing template and `Swing` is its amount. `16th swing` delays every off-16th by up to half a step, `8th swing` every off-8th, `Laid back` everything but the downbeat by up to a fifth

of a step, `Late snare` beats 2 and 4, and `Humanize` delays each step by a repeatable random amount (up to 0.15 of a step) that depends on the step and the cycle, so it replays identically from any position.

- **Rows**: `+ Add Row` under the last row adds a row that starts on the downbeats. `Folder` opens a directory picker; eight random audio files from it (searched recursively; WAV, AIFF, FLAC, OGG, MP3 and whatever else JUCE reads) fill the row's eight sample slots. The folder icon beside the folder's name opens the picker and the reload icon beside it picks eight new files from the same folder; the cross at the row's top-left removes the row. `Rnd` and `Hits` sit on the next line and `M`, `S`, the collapse toggle and the choke box on the last, so the cells take the whole width to the right. `M` mutes and `S` solos: while any row is soloed only soloed rows play, and a soloed row plays even if muted; a silenced row's hit strip is veiled. `-/+` collapses the row to its controls and hit strip and opens it again. The left edge flashes when the row fires.
- **Hit strip**: each cell is a step and draws the sound that step makes: its slot's waveform from its start point, at its speed and pitch, under its attack and decay, over a two-step window, with the slot number in the corner. Click to toggle the step, drag sideways to paint, drag up or down to change its slot, double-click to put the slot back to the row's base, Alt-click to hear the step as it is set, right-click to lock it (a padlock mark; `Rnd` and `Hits` leave a locked cell's hit and slot alone). Right-click locks a lane's bar the same way.
- **Row controls**, above the hit strip, set the row's base sound: `Sample` (1-8), `Attack`, `Decay`, `Tape`, `Pitch`, `Stretch`, `Start`, `Pan`, `Vol`, then the end of the row's chain: `Drive` (a normalised soft clip, 0 dB = clean; a knob only, no lane), `SR` (a sample-and-hold rate reduction, 1-48 kHz, 48 kHz = off), `Bits` (2-16, 16 = off), a 12 dB `HP` and a 12 dB `LP`, and the two sends `Rev` and `Dly` (0-100 %). They are saved with the session but are not host parameters, since rows come and go.
- **Tape, Pitch and Stretch** are three separate controls (and lanes), applied in that order. `Tape` (+/-24 st, a speed of 2^(st/12), so +12 is 2x) is a playback speed that changes pitch and time together, like a sampler or a tape machine. `Pitch` (+/-24 st) then shifts the pitch alone. Both snap to whole semitones, knob and lane alike. `Stretch` (0.25x-4x) then changes the time alone. The playhead moves at Tape x Stretch and the waveform is read at Tape x 2^(Pitch/12); when those agree the sample is simply resampled, otherwise two overlapping 25 ms Hann grains read the waveform at the read rate while the playhead advances, each new grain sliding a little to where its waveform lines up with the grain already playing (WSOLA) so the overlap stays in phase. The first grain starts at its window's peak, so the attack of a hit is untouched. Expect the usual granular artifacts on strong slow-downs: a later transient can land early or repeat by up to a grain.
- **Decay** is a length from -1 to 1, relative to the step at the current tempo and rate: the hit's level follows an S-curve (a raised cosine) from full to silence over 2^(4v) steps, so it holds near full at first, falls fastest in the middle and lands softly. 0 is exactly one step, so the hit hands over cleanly to the next; 0.25 is two steps, 0.5 four, 0.75 eight, and from 0.98 up it is `Full` (the sample plays out). Down shortens it: -0.25 is half a step, -0.5 a quarter, -1 a sixteenth. The knob reads in steps ("4.0 steps", "1/4 step"); the Decay lane's cells draw the step's envelope over one step of time.
- **Retrig**: how many times a step's hit fires within the step, 1 to 8, spread evenly from the hit to the next step boundary (x4 is four hits a quarter of a step apart). A knob and a lane; the lane's cells draw one tick per hit. Every retrigger plays the step's full settings; the row's four voices mean long decays get choked by the later hits.
- **Prob**: the chance (0-100 %) that a step with a hit fires, knob and lane (a full bar is the knob's value, an empty one never). The draw comes from the Prob lane's seed, the step and the cycle, so a given song position always decides the same way, and the lane's Repeat folds it like Evolve.
- **Rvs**: forward or backwards per step, knob and lane. A reversed hit reads the sample from its end towards its start, so `Start` then measures from the end; Tape, Pitch and Stretch work as usual.
- **Lanes**: automation-style lanes per row, shown when the row is open: Attack, Decay, Tape, Pitch, Stretch, Start, Pan, Rvs, Pan, Vol, SR, Bits, HP, LP, Rev, Dly, Retrig, Prob, Chk. A lane is a row of bars, one per step. Decay, Tape, Pitch, Stretch and Pan are bipolar: the centre line is the knob's value and bars go both ways (a

Decay bar is in Decay's own units, so a bar of 0.5 from a knob at 0 is two steps). Attack, Start, HP, Rev, Dly and Retrig are one-sided: a bar grows from the knob's value up to the top of the range. LP, SR, Bits, Prob and Vol are one-sided the other way: a full bar (the default) is the knob's value, open or at level, and an empty bar is 40 Hz, 1 kHz, 2 bits or silence, so Vol works like a velocity lane. Hover a bar for its value in a small readout. Drag to draw (sideways painting works), double-click a bar to clear it, hover for the resolved value. Drive, SR, Bits, HP, LP, Rev and Dly lanes change the row's chain on every step, hit or not, through 5 ms ramps so the change does not click. Each row shows all its lanes or none: the `-/+` button next to M and S opens and closes them.

- **Rnd**: the randomize controls are boxes that show an amount (0-100 %): drag one up or down to set the amount, click it to randomize with that amount. Every lane has an `R` box that moves each unlocked bar by a random amount within +/- the amount. The row has two: `Rnd` changes which sample slot each unlocked step plays (by up to the amount, wrapping), and `Hits` toggles each unlocked step with the amount as the probability.
- **Evolve**: `E` on a lane opens its evolve sublane, a 0…1 amount per step. On every cycle the step's bar moves by a random value within +/- that amount, drawn from a hash of the row's seed, the lane, the step and the cycle number taken from the song position. It is not cumulative: each cycle deviates from the drawn bar, so the same cycle always plays the same way and jumping around the song is safe. The lane shows the evolved value as a red outline over the drawn bar. Each lane has its own seed and `Repeat`, shown in its evolve sublane; typing a seed replays a wander you liked. Repeat folds the cycle number so that lane's deviations come round again every 1-256 cycles of the row (`off` draws a new one on every bar).
- **FX buses** (in the header, after the clock controls): a reverb and a tempo-synced delay sit outside every row's chain. Each row sends to them after its Drive, HP and LP with its `Rev` and `Dly` knobs and lanes; like the filters, the sends follow their lanes on every step. The buses return at unity, so the sends set how much is heard. The reverb has `Size` and `Damp`; the delay has a `Time` in note values (1/32 to 1/2, with triplets and dotted), a `Mode` (`Stereo`: each side repeats on its own side; `Ping-pong`: the input is centred and each repeat crosses over, starting left; `Spin`: the repeats are panned a sixth of a cycle further round each time, so a hit wanders left, centre, right and back), `Feedback`, a `Tone` low-pass in the feedback loop. Since their levels are set by the sends alone, a bus with no sends is silent.
- **Roll** and **Undo** (header): Roll randomizes every row at once, its samples, hits and every lane, each with the amount set in that row's own boxes; Undo puts every row's pattern back as it was before the last Roll.
- **Patterns**: every row holds four patterns, A to D, of hits, locks, lane bars and evolve amounts. The `Pattern` chooser in the header (a host parameter that hosts do not automate, like every choice) picks which one every row plays and shows; `Copy` under it copies each row's current pattern into its other three. Row controls, seeds, Steps and the samples are shared by all four. Older sessions load their one pattern as A with B to D as copies.
- **Fill**: a momentary button in the header (and a host parameter, so it can be mapped or automated). While it is held, every lane's Evolve draws from an alternate seed and every hit retriggers twice as much (at least twice), so a bar becomes a fill and snaps back when released.
- **Choke groups**: the `chk` box sets the row's group, 1 to 4 (`-` = none), and the `Chk` toggle lane says which of its hits do the cutting (the knob is the default, on). When a hit with Chk on fires, every other row in that group is cut within 2 ms, so a closed hat chokes an open hat, while a hit with Chk off lets the group ring.
- **Toggle lanes** (Rvs, Chk) show on or off per cell instead of a bar: click a cell to flip it, drag to paint. The knob is the base state and a cell that is on flips it, so with the Chk knob at `cut` an on cell means that hit does not cut. Rnd on a toggle lane flips each unlocked cell with the amount as the probability.
- Each row has four voices; a fifth hit while four are sounding steals the oldest.

- State recall saves the rows, their steps, slot shifts, lanes, evolve amounts, seed, repeat, controls, mute, the folder and the eight file paths; files that have moved are shown as missing and stay silent. Each file is also saved relative to its row's folder and the folder relative to the home folder, and those are tried first, so the same folders under another user or machine are found. Choices are saved by a stable ID beside their position and lanes with the list of lane IDs, so a session keeps its settings when a list changes; sessions from 1.0.0 load by position as before.
- Host automation: the knobs and Fill can be automated; the choices (Clock, Rate, Groove, Pattern, delay Time and Mode) cannot, since a host stores a choice as its place in the list and that moves when the list grows.

Every parameter

Values and defaults are taken from the plugin. A control's effect can depend on the selected mode; see the operation notes above. Choices keep their saved identity between versions. Knobs and switches can be automated unless marked otherwise.

Parameter	What it does	Values and default
Clock	Host follows DAW song position. Internal runs from the BPM knob, including while the host transport is stopped.	Host, Internal Default: Host Not host automatable
BPM	Sets the internal clock tempo. Host clock uses the DAW tempo.	20.00 to 300.00 Default: 120.00
Rate	Sets the musical duration of each step.	1/4, 1/4T, 1/8D, 1/8, 1/8T, 1/16D, 1/16, 1/16T, 1/32D, 1/32 Default: 1/16 Not host automatable
Groove	Chooses the timing template applied to the song grid.	16th swing, 8th swing, Laid back, Late snare, Humanize Default: 16th swing Not host automatable
Swing	Sets how strongly the chosen Groove shifts step timing.	0.00 to 100.00 % Default: 0.00
Output	Sets the signal level at this stage. Output is the final listening trim unless the signal-flow notes describe a wet-only output control.	-60.00 to 12.00 dB Default: -0.00
Reverb Size	Changes the size of the shared reverb bus.	0.00 to 100.00 % Default: 55.00
Reverb Damp	Darkens the reverb decay by reducing high-frequency persistence.	0.00 to 100.00 % Default: 50.00
Delay Time	Sets the musical length of the shared delay repeats.	1/32, 1/16, 1/8T, 1/16D, 1/8, 1/4T, 1/8D, 1/4, 1/2 Default: 1/8D Not host automatable

Parameter	What it does	Values and default
Delay Mode	Chooses the shared delay routing: stereo, ping-pong or rotating stereo position.	Stereo, Ping-pong, Spin Default: Stereo Not host automatable
Fill	Hold for a fill: alternate evolve deviations on every lane and doubled retriggers (also a host parameter)	Off to On Default: Off
Pattern	Selects the saved pattern of hits and lanes. Row sounds and sample banks are shared across patterns.	A, B, C, D Default: A Not host automatable
Delay Feedback	Returns part of a signal to its input. More feedback increases the strength or persistence of the selected effect.	0.00 to 95.00 % Default: 40.00
Delay Tone	Sets the frequency boundary of this filter or band. Its mode and slope determine which frequencies it changes.	500.0000000 to 20000.0000000 Hz Default: 6000.0004883

Presets and files

Save the DAW project to save the complete instrument. There is no separate Autoloop preset file or preset manager.

Keep the referenced sample folders with your project backups. Relative folder/sample references are used where possible; an absolute reference is retained for older sessions. If a sample folder moves, choose the new location for that row.

Installation and help

Download the macOS archive from the product page and quit your audio host. Place the AU .component in ~/Library/Audio/Plug-Ins/Components, VST3 in ~/Library/Audio/Plug-Ins/VST3, or CLAP in ~/Library/Audio/Plug-Ins/CLAP. The product page also offers an installer with format selection. Restart or rescan the host. To uninstall, quit the host and remove that product's bundles from those folders; saved sessions and user presets are separate.

For questions or a problem report, use BRK Audio support on GitHub. Include the plugin version, macOS version, host and format, sample rate and a short description of how to reproduce it.